Third-level Clifford families with unbounded controlled levels

Abstract

We construct explicit quantum gates of exact Clifford-hierarchy level three and fixed dyadic Pauli period two whose controlled gates attain every exact level at least five. The number of qubits grows with the controlled level. A two-parameter extension gives exact hierarchy levels and periods, with all Pauli phases retained.

Xu and Wang ask how high the controlled level can be when the number of qubits, the level of the original gate, and its dyadic Pauli period are prescribed [1, Open Problem 1]. Our family gives unbounded controlled levels with original level three and fixed period two, as the number of qubits grows. This is a partial answer: it does not determine the maximum at a fixed qubit count. We use the block-extraction principle of [1, Lemma 2] and the necessary period condition of Anderson and Weippert [2], also stated in [1, Theorem 1]. The proofs below keep the specified Pauli phases explicit.

1 Definitions and the explicit family

On n qubits use the actual Pauli group

$$\mathcal{P}_n = \{aX_xZ_z : a \in \{1,-1,i,-i\},\ x,z \in \mathbb{F}_2^n\}, \quad X_x = \prod_j X_j^{x_j}, \quad Z_z = \prod_j Z_j^{z_j}.$$

The Clifford hierarchy is defined recursively by

$$\mathcal{C}_1(n) = \mathcal{P}_n, \qquad \mathcal{C}_{r+1}(n) = \{U \text{ unitary} : UPU^\dagger \in \mathcal{C}_r(n) \text{ for every } P \in \mathcal{P}_n\}.$$

The exact level $\mathrm{lev}(U)$ is the least positive r containing U, when such a level exists. Define the actual dyadic Pauli period by

$$\mathrm{per}(U) = \min\{a \ge 0 : U^{2^a} \in \mathcal{P}_n\},$$

when this set is nonempty. All phases of U are retained. Its controlled matrix is $\mathrm{C}(U) = \mathrm{diag}_c(I,U)$, on a genuine additional input bit c.

For $s \ge 1$ define two layers on an s-qubit block, numbered from 1:

$$C_1 = H_1 \prod_{j\ge1,\ 2j+1\le s} \mathrm{CNOT}_{2j+1\to 2j}, \tag{1.1}$$

$$C_2 = \mathrm{CZ}_{1,2} \prod_{j\ge1,\ 2j+2\le s} \mathrm{CNOT}_{2j+2\to 2j+1} \quad (s \ge 2), \tag{1.2}$$

and $C_2 = I$ when $s = 1$. Gates in each displayed layer have disjoint supports. Let S swap the two complete s-qubit blocks, and put

$$R = C_2C_1, \quad B = (C_1 \otimes I)S, \quad D = C_2 \otimes C_1, \quad A = BD = (I \otimes C_2)S. \tag{1.3}$$

The selector target, on $2s+1$ qubits, is

$$U_s = \mathrm{diag}_b(B,A), \qquad W_s = \mathrm{C}(U_s). \tag{1.4}$$

Both block registers, the selector b, and the outer control c range over their entire input spaces.

For integers $h \geq 1$ and $m \geq 2$ set

$$s_h = \begin{cases} 1, & h = 1, \\ 2^{h-2} + 1, & h \geq 2, \end{cases} \qquad \lambda_m = \exp(i\pi/2^{m+1}), \qquad U_{h,m} = \lambda_m U_{s_h}. \tag{1.5}$$

Theorem 1.1. *For every $h \geq 1$ and $m \geq 2$, the unitary $U_{h,m}$ acts on*

$$n_h = \begin{cases} 3, & h = 1, \\ 2^{h-1} + 3, & h \geq 2, \end{cases}$$

qubits and satisfies

$$\operatorname{lev}(U_{h,m}) = 3, \qquad \operatorname{per}(U_{h,m}) = m, \qquad \operatorname{lev}(\mathrm{C}(U_{h,m})) = \max\{h+4, m+2\}. \tag{1.6}$$

Corollary 1.2. *For every integer $r \geq 5$ there is a finite target with exact level three, actual period two, and exact controlled level r. It is $U_{r-4,2}$ in (1.5).*

2 Elementary hierarchy and period facts

We use the correction $[V, P] = VPV^\dagger P^\dagger$. Membership in $\mathcal{C}_r$, for $r \geq 2$, is equivalent to all its Pauli corrections belonging to $\mathcal{C}_{r-1}$, since multiplication by an actual Pauli preserves every hierarchy level. Induction also proves that the hierarchy is nested, that Clifford conjugation preserves every level, and that multiplication on either side by a Clifford preserves $\mathcal{C}_r$ for $r \geq 2$. Unit scalar multiplication preserves these levels as well. Exact levels at least three are therefore preserved by Clifford factors and unit scalars. These are the usual recursive hierarchy properties; controlled-Clifford jump criteria are treated in [1].

If $\operatorname{diag}(V_0, V_1) \in \mathcal{C}_r$, then both blocks belong to $\mathcal{C}_r$. At level one this follows from the block form of an actual Pauli. The induction step applies the previous-level assertion to the conjugations of $I \otimes P$. Tensoring an identity spectator preserves membership in both directions by the same argument.

Lemma 2.1 (Necessary actual period)**.** *If $\mathrm{C}(V) \in \mathcal{C}_r$ with $r \geq 2$, then*

$$V^{2^{r-2}} \in \mathcal{P}_n. \tag{2.1}$$

Also $V, V^\dagger \in \mathcal{C}_{r-1}$.

Proof. The first outer-control correction is

$$[\mathrm{C}(V), X_c] = \operatorname{diag}(V^\dagger, V) \in \mathcal{C}_{r-1}.$$

Block extraction gives the second assertion. Repeated correction by X_c gives $\operatorname{diag}(V^{-2^j}, V^{2^j}) \in \mathcal{C}_{r-1-j}$ for $0 \leq j \leq r-2$. At the last step its blocks are actual Paulis, including their fourth-root phases. This proves (2.1). □

Lemma 2.2. *If D is a Clifford, $D^2 = I$, and $D \notin \mathcal{P}_n$, then $\mathrm{C}(D)$ has exact level three.*

Proof. For a target Pauli P, its image is $\operatorname{diag}(P, DPD)$, a Pauli times a controlled Pauli, and hence Clifford. The image of X_b is $(I \otimes D)X_b$, also Clifford; Z_b is fixed. Since the destination $\mathcal{C}_2$ is a group, this proves the assertion for every full Pauli string and puts $\mathrm{C}(D)$ in $\mathcal{C}_3$. If it were in $\mathcal{C}_2$, Lemma 2.1 would force D to be an actual Pauli. □

Here a controlled actual Pauli is Clifford: a Hermitian Pauli can be Clifford-conjugated to a product of Z operators, and its controlled version is a product of CZ gates. The allowed factor i^k contributes S_b^k on the control. This verifies the phase convention used in Lemma 2.2.

3 The physical Clifford cycle and its signed endpoint

Both C_1 and C_2 are real Hermitian Clifford involutions. Their product satisfies the actual identity

$$C_1 R C_1 = R^{-1}. \tag{3.1}$$

We compute its action on the complete binary Pauli-label space $V = \mathbb{F}_2^{2s}$. In this computation Y_1 denotes the label $X_1 + Z_1$. Let $F_R = I + N$. Direct propagation through (1.1)–(1.2) gives, for $s \geq 2$,

$$\begin{aligned}
&NX_2 = Z_1, && NX_3 = X_2 + Z_1, \\
&NX_i = X_{i-1} && \text{for even } i \geq 4, \\
&NX_i = X_{i-1} + X_{i-2} && \text{for odd } i \geq 5, \\
&NZ_1 = Y_1 + Z_2, && NY_1 = Z_2, \\
&NZ_i = Z_{i+1} && \text{for odd } i \geq 3, \\
&NZ_i = Z_{i+1} + Z_{i+2} && \text{for even } i \geq 2.
\end{aligned} \tag{3.2}$$

Only existing input labels occur; labels past s in these formulas are zero. The X_3 formula is used only if $s \geq 3$. In the ordered basis

$$X_s, \ldots, X_2, Z_1, Y_1, Z_2, \ldots, Z_s$$

every basis vector is sent to the next one with coefficient one, plus a combination of later vectors. Therefore

$$N^{2s} = 0, \qquad N^{2s-1} X_s = Z_s \neq 0, \qquad \ker N = \langle Z_s \rangle. \tag{3.3}$$

The nonzero power has one surviving path of length $2s-1$, using all adjacent steps. For $s = 1$, use the ordered basis Z_1, Y_1; then $NZ_1 = Y_1$, $NY_1 = 0$, and $R = H_1$.

Let $h = \lceil \log_2(2s) \rceil$ and $q = 2^h$. In characteristic two,

$$(I + N)^{2^a} = I + N^{2^a}.$$

Thus the exact dyadic order of the label action is q: its earlier dyadic powers are nonidentity. In particular no earlier dyadic power of R is even a scalar multiple of a Pauli.

We require an actual matrix endpoint as well. An identity label action means the Clifford is proportional to a Pauli: match its X/Z conjugation signs by a Pauli, and the remaining operator commutes with all Paulis. Commutation with all Z strings makes it diagonal, and commutation with all X strings makes all diagonal entries equal. Hence the remaining operator is scalar. Apply this to R^q. Its Pauli label p is fixed by F_R, since R commutes with R^q. For $s \geq 2$, (3.3) gives $p = 0$ or Z_s. Both corresponding ordered Paulis are real Hermitian. Reality and unitarity consequently force

$$R^q \in \{I, -I, Z_s, -Z_s\}, \qquad R^{2q} = I. \tag{3.4}$$

For $s = 1$, $R^2 = I$ directly. The endpoint label also satisfies

$$p \in \operatorname{im} N^{q/2}. \tag{3.5}$$

Indeed $2s > q/2$, and the last vector of the regular chain lies in this image; at $s = 1$ the actual endpoint label is zero. This derives both the endpoint sign and the image condition from the physical circuit. The regular-unipotent Clifford period mechanism and its resource scale are also discussed in [1, Theorem 4, Proposition 9, Appendix A].

4 Exact target level, period, and controlled lower bounds

From the actual involution identities,

$$B^2 = C_1 \otimes C_1, \quad A^2 = C_2 \otimes C_2, \quad B^4 = A^4 = D^2 = I. \tag{4.1}$$

The second factor of D contains the Hadamard H_1. Its computational columns have two nonzero entries, since the other gates in that layer are disjoint permutations. Tensoring with the unitary C_2 preserves a two-entry column witness. Every scalar multiple of a Pauli has at most one nonzero entry in each column. Thus D is not even projectively Pauli, and in particular is not an actual Pauli. Lemma 2.2 and

$$U_s = (I_b \otimes B)\, \mathrm{C}_b(D)$$

show that U_s has exact level three. Its fourth power is I. Its square is not even projectively Pauli, because its $b = 0$ block is $C_1 \otimes C_1$, which has the same multi-entry obstruction. Therefore $\mathrm{per}(U_s) = 2$.

The scalar λ_m has primitive order 2^{m+2}, satisfies $\lambda_m^{2^m} = i$, and obeys

$$\lambda_m^{2^a} \in \{1, -1, i, -i\} \quad \Longleftrightarrow \quad a \geq m. \tag{4.2}$$

The early powers of U_s are not scalar Paulis; for $a \geq 2$ its dyadic power is I. Equation (4.2) consequently proves $\mathrm{per}(\lambda_m U_s) = m$. Unit scalar multiplication preserves the exact target level three.

Put $E = BDB^\dagger$. Direct use of the block swap gives

$$E = C_1 \otimes C_2, \qquad DE = R \otimes R^{-1} =: R_{\mathrm{big}}. \tag{4.3}$$

The qth power of R_{big} is an actual Pauli by (3.4); earlier dyadic powers have nonidentity label action on a block. Thus $\mathrm{per}(R_{\mathrm{big}}) = h$.

Suppose $W = \mathrm{C}(\lambda_m U_s)$ has a finite level r. Block extraction gives $r \geq 3$; applying it to the first control correction gives $r \geq 4$. The two literal corrections by X_c and then X_b give

$$\begin{aligned} H_0 &= \mathrm{diag}_c(\overline{\lambda_m} U_s^\dagger, \lambda_m U_s), \\ [H_0, X_b] &= \mathrm{diag}_c(D, E) \otimes I_b \in \mathcal{C}_{r-2}. \end{aligned} \tag{4.4}$$

The scalars cancel within the selector correction. Removing D, a constant Clifford, is valid because $r - 2 \geq 2$ and gives $\mathrm{C}(R_{\mathrm{big}}) \otimes I_b \in \mathcal{C}_{r-2}$. Lemma 2.1 implies

$$r \geq h + 4. \tag{4.5}$$

Applied directly to $\mathrm{C}(\lambda_m U_s)$, the same lemma and its actual target period imply

$$r \geq m + 2. \tag{4.6}$$

These inequalities exclude every smaller level once a finite upper bound is supplied.

5 Signed corrections for all physical Pauli strings

We now prove the upper bound first for $W_s = \mathrm{C}(U_s)$, with $q = 2^h$ as above. On the two-block state space set

$$T_0 = R \otimes I, \qquad T_1 = I \otimes R, \qquad V_a = \operatorname{im} N^a \oplus \operatorname{im} N^a.$$

The shifts commute, have actual order dividing $2q$, and satisfy

$$BT_0B^{-1} = T_1, \qquad BT_1B^{-1} = T_0^{-1}, \qquad D = T_0B^2. \tag{5.1}$$

The four (c, b) branches of W_s are therefore the actual word

$$F(c, b) = T_1^{cb}B^{c+2cb}, \qquad c, b \in \{0, 1\}. \tag{5.2}$$

All powers here and below are powers of actual unitary matrices, including negative integer powers.

For a state label $v = (x, z)$ let $P_v = X_xZ_z$. The full signed action is computable by the following elementary rules, together with composition:

$$\begin{aligned} H_jP_{x,z}H_j &= (-1)^{x_jz_j}P_{x',z'}, & (x'_j, z'_j) &= (z_j, x_j), \\ \mathrm{CNOT}_{a\to b}P_{x,z}\mathrm{CNOT}_{a\to b} &= P_{x',z'}, & x'_b &= x_b + x_a, \quad z'_a = z_a + z_b, \\ \mathrm{CZ}_{a,b}P_{x,z}\mathrm{CZ}_{a,b} &= (-1)^{x_ax_b}P_{x,z'}, & z'_a &= z_a + x_b, \quad z'_b = z_b + x_a. \end{aligned}$$

All unlisted coordinates are unchanged. Block swap simply swaps the two full labels. Hence every real Clifford occurring in the proof has an actual action

$$FP_vF^\dagger = (-1)^{\kappa_F(v)}P_{L_Fv} \tag{5.3}$$

on every state Pauli. The signs are retained, including those produced by the Hadamard gates. Ordered Paulis obey

$$P_vP_w = (-1)^{z_v\cdot x_w}P_{v+w}, \qquad P_v^\dagger = (-1)^{z_v\cdot x_v}P_v. \tag{5.4}$$

Consider an intermediate diagonal multiplexer whose branches are

$$(-1)^{f(u)}P_{y(u)}F(u), \qquad u = (c, b) \in \mathbb{F}_2^2. \tag{5.5}$$

The functions f and y may be arbitrary. A full input Pauli has arbitrary control translation $a \in \mathbb{F}_2^2$, arbitrary control Z label, arbitrary state label w, and a scalar in $\{1, -1, i, -i\}$. The input scalar and control Z phases cancel in its normalized correction. Put

$$G(u) = F(u)F(u + a)^\dagger.$$

Direct matrix multiplication and (5.4) give its state label

$$y'(u) = y(u) + L_G(u)y(u + a) + L_F(u)w + L_G(u)w. \tag{5.6}$$

For completeness, its real branch sign is explicitly determined as follows. Put

$$v_1 = y(u), \quad v_2 = L_F(u)w, \quad v_3 = L_G(u)y(u + a), \quad v_4 = L_G(u)w.$$

Then the new sign exponent, in $\mathbb{F}_2$, is

$$f'(u) = f(u) + f(u+a) + \kappa_{F(u)}(w) + \kappa_{G(u)}(y(u+a)) + \kappa_{G(u)}(w) + z_{v_3} \cdot x_{v_3} + z_{v_4} \cdot x_{v_4} + \sum_{k<l} z_{v_k} \cdot x_{v_l}. \tag{5.7}$$

Thus arbitrary mixed labels and all allowed input phases are accounted for by actual matrix identities. Subsequent steps remain in the real signed class (5.5).

6 A filtration closed under every Pauli correction

For $0 \le j \le h-1$, consider profiles with

$$\begin{aligned} F(u) &= T_0^{t_0(u)} T_1^{t_1(u)}, \\ t_i(c,b) &= t_{i,c}c + t_{i,b}b + t_{i,12}cb, \qquad (6.1) \\ 2^j &\mid t_{i,c}, t_{i,b}, \qquad 2^{j+1} \mid t_{i,12}, \\ y_{12} &\in V_{2^j}. \qquad (6.2) \end{aligned}$$

Here $y_{12} = y(0,0) + y(1,0) + y(0,1) + y(1,1)$ is the quadratic Boolean coefficient. Integer coefficients are kept unreduced; if a modulus is used it is the actual period $2q$, not the label period q.

Lemma 6.1. *If (6.2) holds at depth $j < h-1$, every full Pauli correction, after removing a constant right Clifford, has the same form at depth $j+1$.*

Proof. For a control translation $a = (a_c, a_b)$, normalizing the constant branch replaces each exponent polynomial $t(u)$ by

$$t(u) - t(u+a) + t(a).$$

Write Boolean addition as $c \oplus a_c = c + a_c - 2ca_c$ over the integers. Expansion gives the new coefficients

$$\begin{aligned} t'_c &= 2a_c t_c - a_b(1-2a_c)t_{12}, \\ t'_b &= 2a_b t_b - a_c(1-2a_b)t_{12}, \qquad (6.3) \\ t'_{12} &= 2(a_c + a_b - 2a_c a_b)t_{12}. \end{aligned}$$

The two linear divisibilities and the deeper quadratic divisibility advance by one power of two.

We verify the label condition as well. Put $k = 2^j$. A shift exponent divisible by k has label action I plus a polynomial in N divisible by N^k on its block, and thus raises the filtration by k. This holds for negative exponents too, since $I + N$ is invertible and its inverse is a finite polynomial in N. For a single block, the quadratic coefficient of the label profile with exponents $\alpha c + \beta b + \gamma cb$ is

$$((I+N)^\alpha - I)((I+N)^\beta - I) + (I+N)^{\alpha+\beta}((I+N)^\gamma - I). \tag{6.4}$$

When $k \mid \alpha, \beta$ and $2k \mid \gamma$, both terms raise the filtration by $2k$. Products between differences supported on separate blocks are zero.

Before constant normalization, the difference profile G has a constant action G_0 with $G_0 - I$ raising by k. Its nonconstant coefficients raise by $2k$, by (6.3) and (6.4). In (5.6), the quadratic coefficient of $y + G_0 y(\cdot + a)$ is $(I + G_0)y_{12} \in V_{2k}$. The other terms involving the nonconstant part of G or the inserted state label w are already in V_{2k}. Thus $y'_{12} \in V_{2k}$. Removing the constant right

Clifford changes the Clifford profile but leaves the left Pauli label unchanged. The sign (5.7) is still an arbitrary actual real sign function. □

Lemma 6.2 (Terminal level)**.** *Every signed profile satisfying* (6.2) *at* $j = h - 1$ *belongs to* $\mathcal{C}_3$.

Proof. Let $k = q/2$. The quadratic exponents are multiples of q, and the linear exponents are multiples of k. By (3.4), each T_i^q is a real Hermitian actual Pauli with label in V_k, and $T_i^{2q} = I$. Absorb the qth-power factors into the left Pauli dressing. This preserves $y_{12} \in V_k$. The remaining profile is

$$F_*(c, b) = S_c^c S_b^b, \tag{6.5}$$

where S_c, S_b are commuting products of T_0^k, T_1^k. Their squares are actual signed Paulis and their fourth powers are I.

Their label differences raise into V_k, annihilate V_k, and have pairwise product zero because $N^q = 0$. Their label matrices are involutions and the label profile (6.5) is affine in c, b. For an X_c correction, the exact Clifford difference is

$$F_*(c, b)F_*(c + 1, b)^{-1} = S_c^{-1}(S_c^2)^c.$$

It is a constant Clifford times a linear controlled actual Pauli; the X_b formula is identical with S_b. These corrections have constant label action, which fixes V_k. Equation (5.6) therefore has zero quadratic coefficient, since $(I + L_G)y_{12} = 0$ and $L_F w$ is affine. For a state X or Z generator, $a = 0$; then (5.6) reduces to $L_F w + w$, again affine. Control Z generators commute with the multiplexer.

An affine Pauli-label dressing is Clifford: factor it into a constant Pauli and two controlled Paulis. The resulting product signs are functions of only c, b. Every real sign function on two bits is a product of a constant sign, Z_c, Z_b, and $\mathrm{CZ}_{c,b}$, so it too is Clifford. Equations (5.7) and (6.5) now show that each generator correction is Clifford. The group property of $\mathcal{C}_2$ extends this to every full Pauli. This generator argument is confined to this $\mathcal{C}_3$ terminal step. □

Backward induction using Lemmas 6.1 and 6.2 proves

$$\text{every depth-}j\text{ profile is in } \mathcal{C}_{h-j+2}. \tag{6.6}$$

At each backward step the removed right factor is Clifford and the destination level is at least three, so the hierarchy factor rule applies.

7 Entry of the circuit after two corrections

We show that two arbitrary full Pauli corrections of (5.2), allowing zero control translations and arbitrary state labels, enter depth zero. Define the normalized profile difference

$$\mathcal{D}_a F(u) = F(u)F(u + a)^{-1}F(a).$$

The last factor is a constant Clifford, so this normalization preserves membership at all levels used below. Write $a = (a_c, a_b)$ and treat its bits as integers in the exponent formulas. Direct substitution of the four branches of (5.2) into (5.1) gives

$$\mathcal{D}_a F(c, b) = T_0^{t_0(c,b)} T_1^{t_1(c,b)} K^{(a_c+a_b)c+a_c b}, \qquad K = B^2, \tag{7.1}$$

where the exponent of the actual involution K is binary, and

$$t_0(c,b) = -a_c a_b c + a_c b + a_c(2a_b - 1)cb,$$
$$t_1(c,b) = (1 - a_c)a_b c + a_c cb. \tag{7.2}$$

Also $K^2 = I$ and $KT_iK^{-1} = T_i^{-1}$ for $i = 0, 1$.

For any second translation $d = (d_c, d_b)$ put

$$\delta = (-1)^{(a_c+a_b)d_c + a_c d_b} \in \{1, -1\}.$$

A second normalized difference of (7.1) cancels K and has shift exponents

$$t_i(u) - \delta t_i(u + d) + \delta t_i(d). \tag{7.3}$$

For a polynomial $t = t_c c + t_b b + t_{12} cb$ the new quadratic coefficient is

$$\big[1 - \delta(1 - 2d_c)(1 - 2d_b)\big]t_{12}, \tag{7.4}$$

which is even for either value of δ. The new linear coefficients are integers. Thus (6.2)'s exponent condition at depth zero holds without a phase quotient.

It remains to include the Pauli dressings generated by those same two arbitrary corrections. From (3.1), C_1 preserves every $\operatorname{im} N^a$: conjugating N by C_1 gives $(I + N)^{-1} - I = N(I + N)^{-1}$. Its action on the one-dimensional quotient $V/\operatorname{im} N$ is the identity. On the two-block head quotient V_0/V_1, B consequently swaps the heads, with action J satisfying $J^2 = I$, while D and both shifts act trivially. The original profile (5.2) has head action J^c.

Starting from zero dressing, (5.6) gives on this quotient, for the first state label w,

$$y_1(c,b) = J^c w + J^{a_c} w.$$

This is affine in c and independent of b. The first normalized Clifford profile (7.1) has identity head action: the shifts and $K = B^2$ all act trivially there. Its difference profile also has identity head action. Hence for any second state label w', (5.6) reduces on the head quotient to

$$y_2(u) = y_1(u) + y_1(u + d) + w' + w' = y_1(u) + y_1(u + d),$$

which is constant. Its quadratic coefficient therefore lies in V_1. Constants removed during normalization only conjugate subsequent arbitrary state Paulis by fixed Cliffords, and those Paulis still range over the entire group. All sign factors remain those of (5.7).

We have proved entry into depth zero for every pair of full Pauli corrections. By (6.6), the second corrections belong to $\mathcal{C}_{h+2}$. Recursing upward twice proves

$$W_s \in \mathcal{C}_{h+4}. \tag{7.5}$$

The proof also covers $h = 1, s = 1$: then $q = 2$, depth zero is already terminal, and the actual $R^2 = I$ supplies its endpoint. No initialized register or restricted input subspace is used.

8 The actual scalar extension and exactness

Lemma 8.1 (Branch phase)**.** *Suppose* $V = \operatorname{diag}(V_0, V_1) \in \mathcal{C}_r$, $r \geq 2$, *and* $|z| = 1$. *If* $z^{2^a} \in \{1, -1, i, -i\}$, *then*

$$\operatorname{diag}(V_0, zV_1) \in \mathcal{C}_{\max\{r, a+2\}}. \tag{8.1}$$

Proof. Induct on a. For $a = 0$ the phase factor is a controlled actual scalar Pauli, hence Clifford. If $r = 2$, factor out the one-bit phase $\mathrm{diag}(1, z)$. Its repeated X corrections square its relative phase, so $z^{2^a} \in \{1, -1, i, -i\}$ puts it in $\mathcal{C}_{a+2}$. Multiplication by the original Clifford V preserves that level.

For $r \geq 3$, take any full input Pauli and its correction. If its control X bit is zero, the branch scalar cancels and the correction is unchanged. If that bit is one, direct block multiplication gives, up to the unit scalar $\bar{z}$, a correction of the form $\mathrm{diag}(M, z^2 N)$, where $\mathrm{diag}(M, N)$ is the original correction in $\mathcal{C}_{r-1}$. Apply the induction hypothesis with $a - 1$, $r - 1$, and z^2. This gives level at most $\max\{r - 1, a + 1\}$ for that full correction. The common unit scalar is immaterial at this level, which is at least two. Control Z components and actual input scalar phases cancel in the same matrix identity. Recursion proves (8.1). □

For the chosen s_h, the inequality $2^{h-1} < 2s_h \leq 2^h$ shows that (7.5) uses precisely the prescribed h in (1.5). Lemma 8.1, with $z = \lambda_m$ and $a = m$, yields

$$\mathrm{C}(U_{h,m}) \in \mathcal{C}_{\max\{h+4, m+2\}}.$$

The actual lower bounds (4.5) and (4.6) prove exactness. The target and period assertions were proved in Section 4; the qubit count is $2s_h + 1$. This establishes Theorem 1.1 and, by $h = r - 4, m = 2$, Corollary 1.2.

References

[1] Y. Xu and X. Wang. *Controlled jump in the Clifford hierarchy.* arXiv:2602.22201v1, 2026.

[2] J. T. Anderson and M. Weippert, *Controlled gates in the Clifford hierarchy* (2024). arXiv:2410.04711v2.